CS-338 Intro to the Theory of Computation

- Lecture #15

➢ **Complexity Theory: Cook-Levin Theorem**

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Overview

Last time:
- NP-completeness
- $3SAT \leq_P CLIQUE$
- $3SAT \leq_P HAMPATH$

Today: (Sipser §7.4)
- Cook-Levin Theorem: SAT is NP-complete
- $3SAT$ is NP-complete

Quick Review

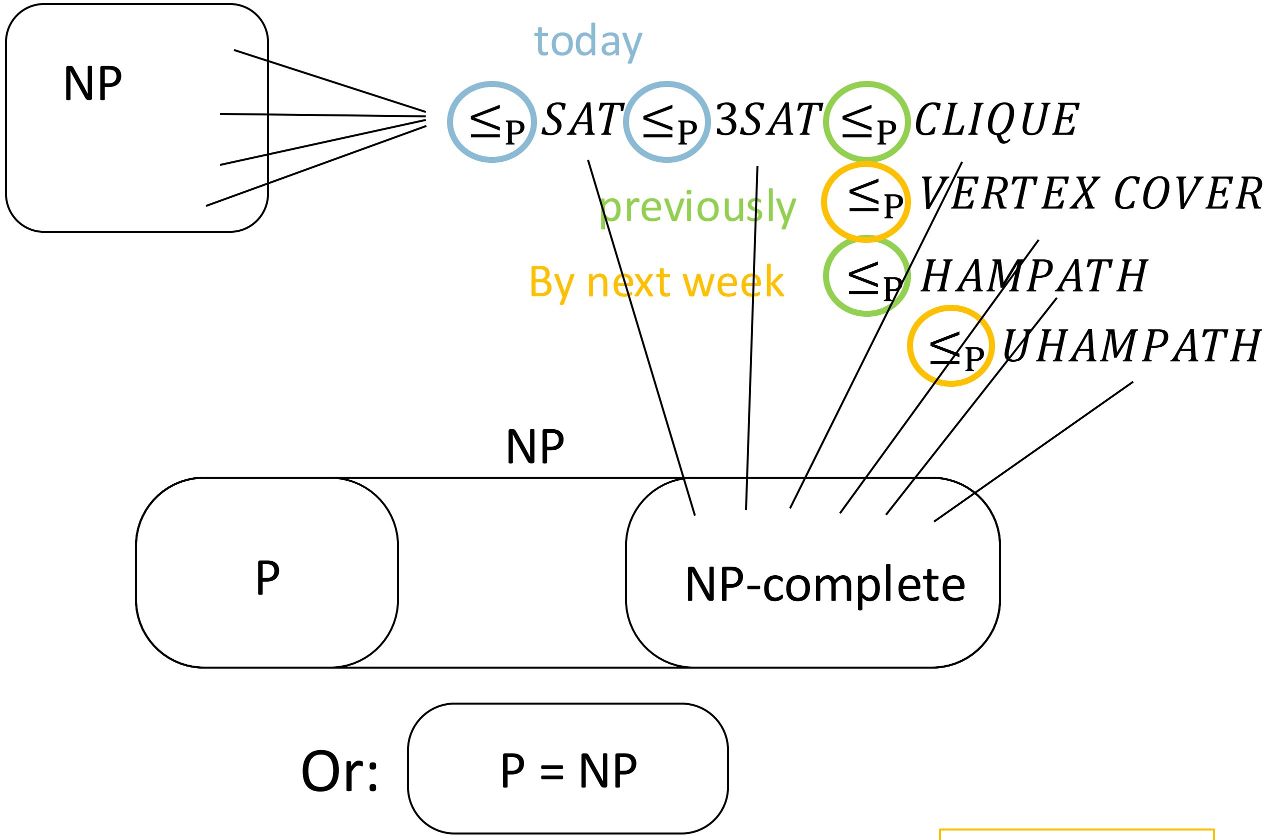
Defn: B is <u>NP-complete</u> if
1) $B \in$ NP
2) For all $A \in$ NP, $A \leq_{\mathrm{P}} B$

If B is NP-complete and $B \in$ P then P $=$ NP.

Importance of NP-completeness
1) Evidence of computational intractability.
2) Gives a good candidate for proving P $\neq$ NP.

To show some language C is NP-complete,
show $3SAT \leq_{\mathrm{P}} C$.

or some other previously shown
NP-complete language



Check-in 15.1

Cook-Levin Theorem (idea)

Theorem: SAT is NP-complete

Proof: 1) $SAT \in NP$ (done)

2) Show that for each $A \in NP$ we have $A \leq_P SAT$:

Let $A \in NP$ be decided by NTM M in time n^k.

Give a polynomial-time reduction f mapping A to SAT.

$f: \Sigma^* \rightarrow$ formulas

$f(w) = \langle \phi_{M,w} \rangle$

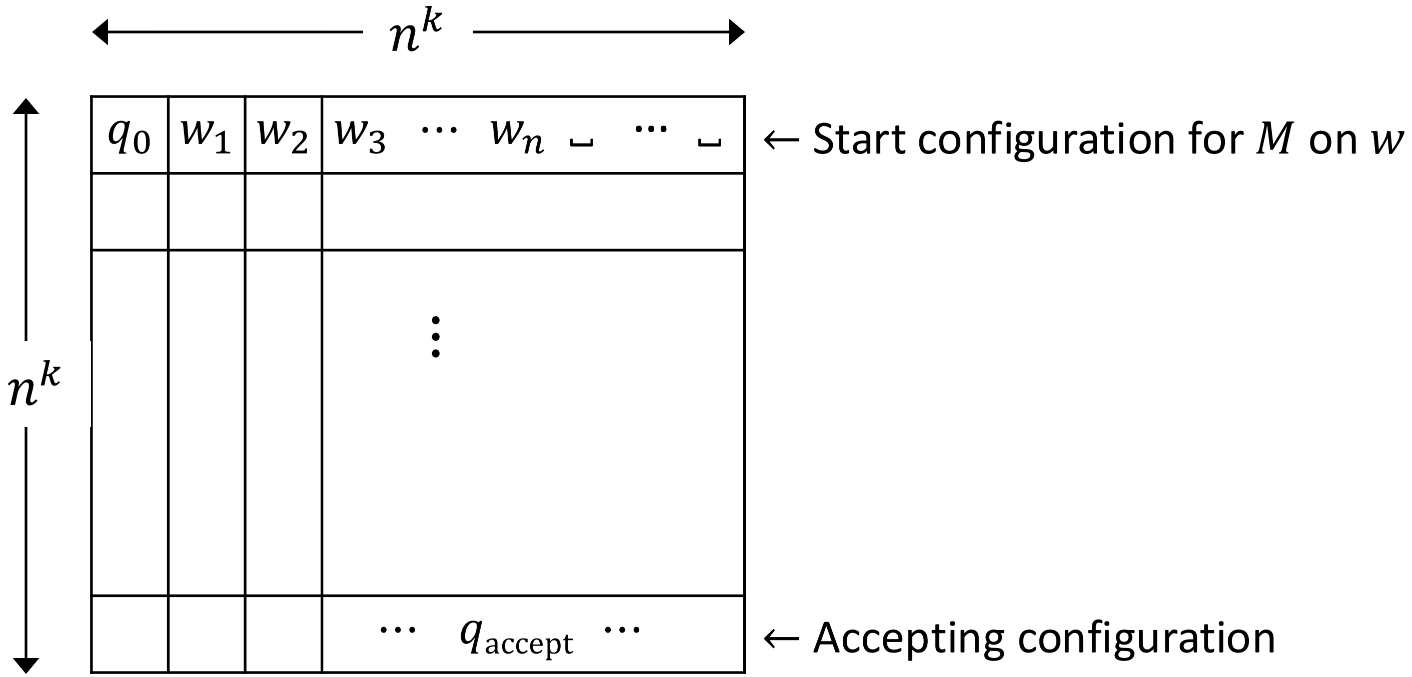
$w \in A$ iff $\phi_{M,w}$ is satisfiable

Idea: $\phi_{M,w}$ simulates M on w. Design $\phi_{M,w}$ to "say" M accepts w.

Satisfying assignment to $\phi_{M,w}$ is a computation history for M on w.

Tableau for M on w

Defn: An (accepting) tableau for NTM M on w is an $n^k \times n^k$ table representing an computation history for M on w on an accepting branch of the nondeterministic computation.

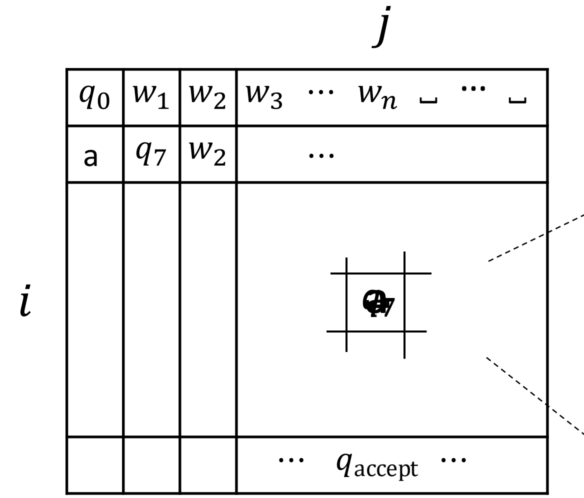


Construct $\phi_{M,w}$ to "say" M accepts w.

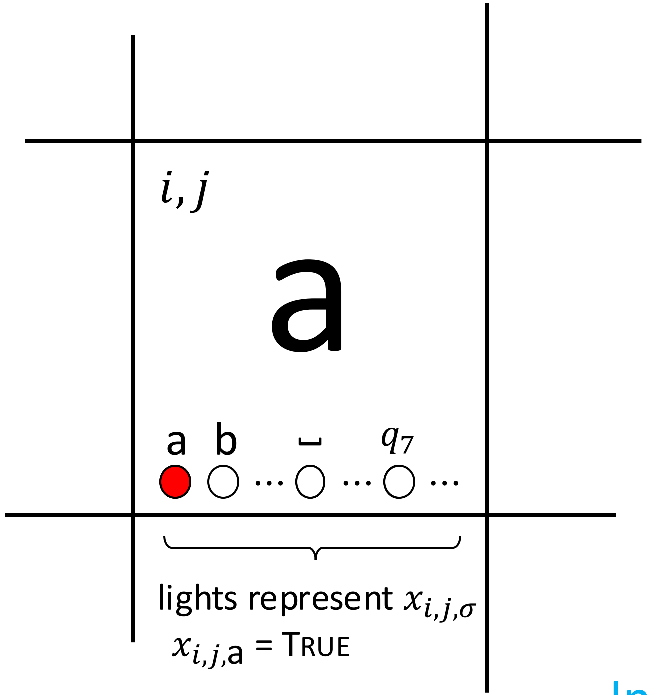
$\phi_{M,w}$ "says" a tableau for M on w exists.

$\phi_{M,w} = \phi_{\text{cell}} \wedge \phi_{\text{start}} \wedge \phi_{\text{move}} \wedge \phi_{\text{accept}}$

Constructing $\phi_{M,w}$: ϕ_{cell}



Cell i,j can contain any symbol in $\Gamma \cup Q$



The variables of $\phi_{M,w}$ are $x_{i,j,\sigma}$ for $1 \leq i,j \leq n^k$ and $\sigma \in \Gamma \cup Q$.

$x_{i,j,\sigma}$ = TRUE means cell i,j contains σ.
(INDICATOR VARIABLE)

Check-in 16.2

How many variables does $\phi_{M,w}$ have?
Recall that $n = |w|$.

(a) $O(n)$
(b) $O(n^2)$
(c) $O(n^k)$
(d) $O(n^{2k})$

$\phi_{M,w}$ "says" a tableau for M on w exists.

$$\phi_{M,w} = \phi_{\text{cell}} \wedge \phi_{\text{start}} \wedge \phi_{\text{move}} \wedge \phi_{\text{accept}}$$

ϕ_{cell} "says" exactly one light is on per cell
i.e., exactly one $x_{i,j,\sigma}$ = TRUE for each i,j.

In every cell at least one light and at most one light

$$\phi_{\text{cell}} = \bigwedge_{1 \leq i,j \leq n^k} \left(\overbrace{\bigvee_{\sigma \in \Gamma \cup Q} x_{i,j,\sigma}}^{x_{i,j,\sigma_1} \vee x_{i,j,\sigma_2} \vee \cdots \vee x_{i,j,\sigma_t}} \wedge \bigwedge_{\substack{\sigma,\tau \in \Gamma \cup Q \\ \sigma \neq \tau}} \left(\overline{x_{i,j,\sigma} \wedge x_{i,j,\tau}} \right) \right)$$

Check-in 16.2

Constructing $\phi_{M,w}$: ϕ_{start} and ϕ_{accept}

	1	2	3	…… n^k	
1	q_0	w_1	w_2	w_3 $\cdots$ w_n $\sqcup$ $\cdots$ $\sqcup$	← Start configuration
	a	q_7	w_2	$\cdots$	
n^k				$\cdots$ q_{accept} $\cdots$	← Accepting configuration

$\phi_{M,w}$ "says" a tableau for M on w exists.

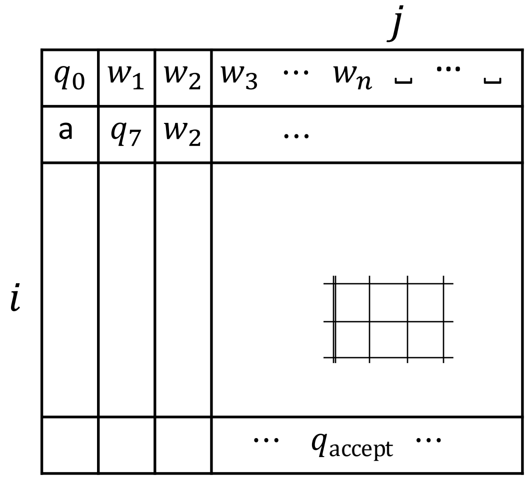
$\phi_{M,w} = \phi_{\text{cell}} \wedge \phi_{\text{start}} \wedge \phi_{\text{move}} \wedge \phi_{\text{accept}}$

ϕ_{cell} done ✓

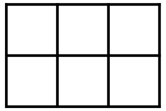
$\phi_{\text{start}} = x_{1,1,q_0} \wedge x_{1,2,w_1} \wedge x_{1,3,w_2} \wedge \cdots \wedge x_{1,n^k,\sqcup}$

$$\phi_{\text{accept}} = \bigvee_{1 \le j \le n^k} x_{n^k,j,q_{\text{accept}}}$$

Constructing $\phi_{M,w}$: ϕ_{move}



2×3 neighborhood



Legal neighborhoods: consistent with M's transition function

potential examples:

a	q_7	b
q_3	a	c

a	b	c
a	b	c

a	b	c
a	b	q_5

a	b	c
d	b	c

Illegal neighborhoods: not consistent with M's transition function

examples:

a	b	c
a	d	c

a	b	c
a	q_2	c

a	q_7	c
a	b	c

a	q_7	c
q_3	d	q_4

Claim: If every 2×3 neighborhood is legal then tableau corresponds to a computation history.

$\phi_{M,w}$ "says" a tableau for M on w exists.

$\phi_{M,w} = \phi_{\text{cell}} \wedge \phi_{\text{start}} \wedge \phi_{\text{move}} \wedge \phi_{\text{accept}}$

✓ ✓ ✓

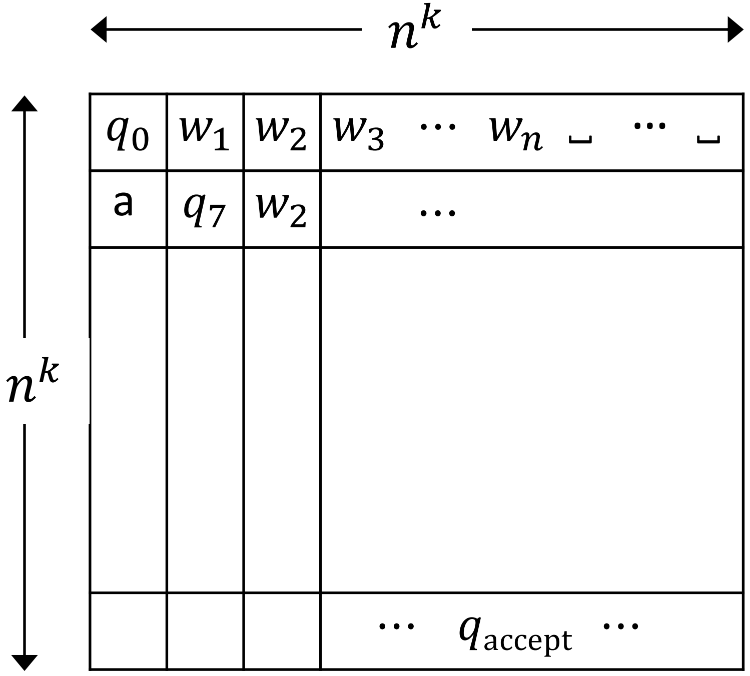
The (i,j) window has has cell[i,j] in the **upper centre** position:

1) If in the **upper** configuration every cell contains a tape symbol then the centre symbol must appear the same in the lower config.
2) If the state symbol appears in the centre top of the **upper** config. then the transition rules will ensure the move is legal.

$$\phi_{\text{move}} = \bigwedge_{1<i,j<n^k} \left(\bigvee_{\substack{\text{Legal} \\ \begin{array}{|c|c|c|}\hline r & s & t \\ \hline v & y & z \\ \hline\end{array}}} \left(x_{i,j-1,r} \wedge x_{i,j,s} \wedge x_{i,j+1,t} \wedge x_{i+1,j-1,v} \wedge x_{i+1,j,y} \wedge x_{i+1,j+1,z} \right) \right)$$

Says that the neighborhood at i, j is legal

Conclusion: SAT is NP-complete



Summary:

For $A \in$ NP, decided by NTM M,
we gave a reduction f from A to SAT:

f: $\Sigma^* \;\rightarrow\;$ formulas

$f(w) \;=\; \langle \phi_{M,w} \rangle$

$w \in A$ iff $\phi_{M,w}$ is satisfiable.

$\phi_{M,w} = \phi_{\text{cell}} \wedge \phi_{\text{start}} \wedge \phi_{\text{move}} \wedge \phi_{\text{accept}}$

The size of $\phi_{M,w}$ is roughly the size of the tableau for M on w, so size is $O\left(n^k \times n^k\right) = O\left(n^{2k}\right)$.

Therefore f is computable in polynomial time.

$3SAT$ is NP-complete

Theorem: $3SAT$ is NP-complete

Proof: Show $SAT \leq_P 3SAT$

Give reduction f converting formula ϕ to 3CNF formula ϕ', <u>preserving satisfiability</u>.
(Note: ϕ and ϕ' are not logically equivalent)

a	b	a ∧ b = c	
1	1	1	$(a \wedge b) \rightarrow c$
0	1	0	$(\bar{a} \wedge b) \rightarrow \bar{c}$
1	0	0	$(a \wedge \bar{b}) \rightarrow \bar{c}$
0	0	0	$(\bar{a} \wedge \bar{b}) \rightarrow \bar{c}$

Assumes:

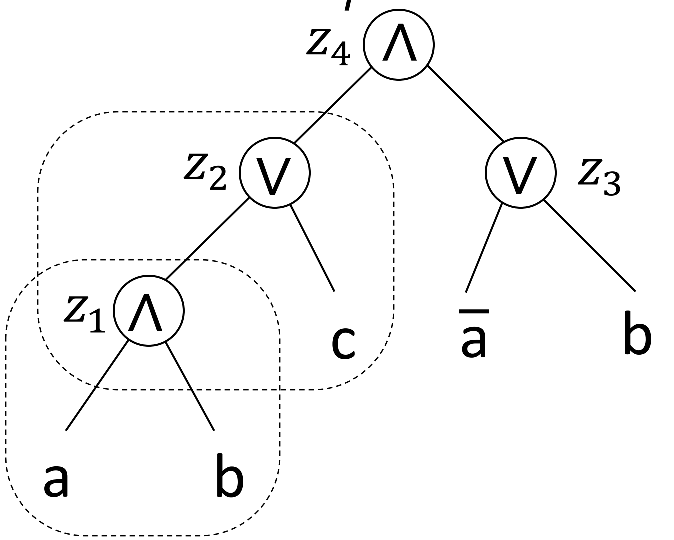
- NOT operations do not appear

(negated variables are pushed to the leaves)

- Unambiguous parenthesisation

Example: Say $\phi = \big((a \wedge b) \vee c\big) \wedge (\bar{a} \vee b)$

Tree structure for ϕ:



Logical equivalence: $(A \rightarrow B)$ and $(\bar{A} \vee B)$ $\overline{(A \wedge B)}$ and $(\bar{A} \vee \bar{B})$

$$\phi' = \big((a \wedge b) \rightarrow z_1\big) \wedge \big((\bar{a} \wedge b) \rightarrow \bar{z}_1\big) \wedge \big((a \wedge \bar{b}) \rightarrow \bar{z}_1\big) \wedge \big((\bar{a} \wedge \bar{b}) \rightarrow \bar{z}_1\big)$$

$$\wedge \big((z_1 \wedge c) \rightarrow z_2\big) \wedge \big((\bar{z}_1 \wedge c) \rightarrow z_2\big) \wedge \big((z_1 \wedge \bar{c}) \rightarrow z_2\big) \wedge \big((\bar{z}_1 \wedge \bar{c}) \rightarrow \bar{z}_2\big)$$

⋮ repeat for each z_i

$$\wedge \ (z_4 \vee z_4 \vee z_4)$$

Observe that $\big((a \wedge b) \rightarrow c\big)$ is logically equivalent to $(\bar{a} \vee \bar{b} \vee c)$

$$\Big[\big((a \wedge b) \rightarrow c\big) \leftrightarrow \big(\overline{(a \wedge b)} \vee c\big) \leftrightarrow \big((\bar{a} \vee \bar{b}) \vee c\big) \leftrightarrow (\bar{a} \vee \bar{b} \vee c)\Big]$$

Check-in 16.3

$3SAT$ is NP-complete: Direct Proof

Theorem: $3SAT$ is NP-complete

Proof: Modify proof that SAT is NP-complete to produce a 3CNF formula

$$\phi_{M,w} = \phi_{\text{cell}} \wedge \phi_{\text{start}} \wedge \phi_{\text{move}} \wedge \phi_{\text{accept}}$$

$$\phi_{\text{cell}} = \bigwedge_{1\le i,j\le n^k} \left(\bigvee_{\sigma\in\Gamma\cup Q} x_{i,j,\sigma} \quad \bigwedge_{\substack{\sigma,\tau\in\Gamma\cup Q \\ \sigma\neq\tau}} \left(\overline{x_{i,j,\sigma} \wedge x_{i,j,\tau}} \right) \right)$$

$$\phi_{\text{start}} = x_{1,1,q_0} \wedge x_{1,2,w_1} \wedge x_{1,3,w_2} \wedge \cdots \wedge x_{1,n^k,\sqcup}$$

$$\phi_{\text{accept}} = \bigvee_{1\le j\le n^k} x_{n^k,j,q_{\text{accept}}}$$

$$\phi_{\text{move}} = \bigwedge_{1<i,j<n^k} \left(\bigvee_{\text{Legal}} \left(x_{i,j-1,r} \wedge x_{i,j,s} \wedge x_{i,j+1,t} \wedge x_{i+1,j-1,v} \wedge x_{i+1,j,y} \wedge x_{i+1,j+1,z} \right) \right)$$

r	s	t
v	y	z

Eg.,

$$(x_1 \vee \cdots \vee x_k \wedge (\overline{x_1 \wedge x_2})) \wedge (x_l \vee \cdots \vee x_p \wedge (\overline{x_l \wedge x_m}))$$

$$= (x_1 \vee \cdots \vee x_k) \wedge (\overline{x_1 \wedge x_2}) \wedge (x_l \vee \cdots \vee x_p) \wedge (\overline{x_l \wedge x_m})$$

$$= (x_1 \vee \cdots \vee x_k) \wedge (\overline{x_1} \vee \overline{x_2}) \wedge (x_l \vee \cdots \vee x_p) \wedge (\overline{x_l} \vee \overline{x_m})$$

This one is in 1-CNF

This one is just one big OR clause => in CNF

1. We want to check that they are or convert into CNF
2. Then convert into 3CNF

$3SAT$ is NP-complete: Direct Proof (2)

Theorem: $3SAT$ is NP-complete

Proof: Modify proof that SAT is NP-complete to produce a 3CNF formula

$$\phi_{M,w} = \underset{\checkmark}{\phi_{\text{cell}}} \wedge \underset{\checkmark}{\phi_{\text{start}}} \wedge \phi_{\text{move}} \wedge \underset{\checkmark}{\phi_{\text{accept}}}$$

1. We want to check that they are or convert into CNF
2. Then convert into 3CNF

$$\phi_{\text{move}} = \bigwedge_{1<i,j<n^k} \left(\bigvee_{\text{Legal}} \left(x_{i,j-1,r} \wedge x_{i,j,s} \wedge x_{i,j+1,t} \wedge x_{i+1,j-1,v} \wedge x_{i+1,j,y} \wedge x_{i+1,j+1,z} \right) \right)$$

r	s	t
v	y	z

$(x_1 \wedge x_2 \wedge x_3) \vee (x_4 \wedge x_5 \wedge x_6)=$

$= (x_1 \vee (x_4 \wedge x_5 \wedge x_6)) \wedge (x_2 \vee (x_4 \wedge x_5 \wedge x_6)) \wedge (x_3 \vee (x_4 \wedge x_5 \wedge x_6))$

$= (x_1 \vee x_4) \wedge (x_1 \vee x_5) \wedge (x_1 \vee x_6) \wedge (x_2 \vee (x_4 \wedge x_5 \wedge x_6)) \wedge \ldots$

This is an OR of ANDs so it is more complicated to convert.

- We use the **distributive** laws for AND and OR:

- $P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$
- $P \vee (Q \wedge R) = (P \vee R) \wedge (P \vee R)$

In general this conversion formula grows exponentially, but it will only be exponential in the size of the window (**6**) !!!

$3SAT$ is NP-complete: Direct Proof (3)

Theorem: $3SAT$ is NP-complete

Proof: Modify proof that SAT is NP-complete to produce a 3CNF formula

$$\phi_{M,w} = \underset{\checkmark}{\phi_{\text{cell}}} \wedge \underset{\checkmark}{\phi_{\text{start}}} \wedge \underset{\checkmark}{\phi_{\text{move}}} \wedge \underset{\checkmark}{\phi_{\text{accept}}}$$

1. We want to check that they are or convert into CNF ✓
2. Then convert into 3CNF

Now for the final conversion:

If the clause has:

- 1 literal: we add two "dummy" literals
  - Eg. $x_1 \rightarrow (x_1 \vee x_1 \vee x_1)$
- 2 literals: we add one "dummy" literal
  - Eg. $(x_1 \vee x_2) \rightarrow (x_1 \vee x_2 \vee x_1)$
- 3 literals: happy ☺
- 4 literals: we split it into several clauses and add extra variables to preserve the logic
  - Eg. $(x_1 \vee x_2 \vee x_3 \vee x_4) \rightarrow (x_1 \vee x_2 \vee z) \wedge (x_3 \vee x_4 \vee \bar{z})$ where z is a new variable
  - If the original clause is satisfied, then we can also satisfy the new one by setting z appropriately
- l literals **(in general)**: we can replace it with a CNF formula by adding $l-2$ clauses: $(x_1 \vee x_2 \vee \cdots \vee x_l) \rightarrow$

$$(x_1 \vee x_2 \vee z_1) \wedge (\overline{z_1} \vee x_3 \vee z_2) \wedge (\overline{z_2} \vee x_4 \vee z_3) \wedge \cdots \wedge (\overline{z_{l-3}} \vee x_{l-1} \vee x_l)$$

Quick review of today

1. SAT is NP-complete
2. $3SAT$ is NP-complete